

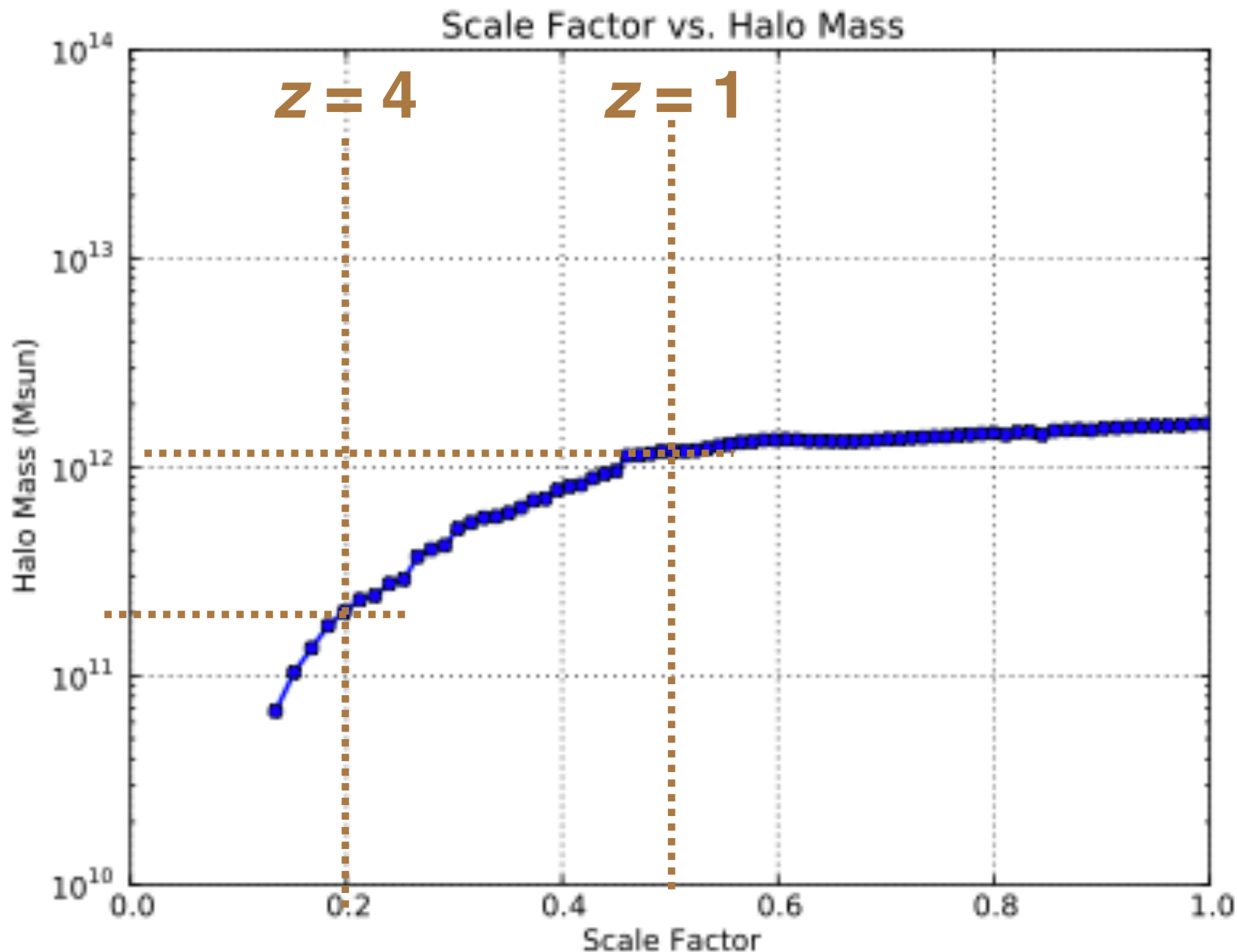
**Expected Stellar Mass
at $z = 4$ for AGORA
halo**

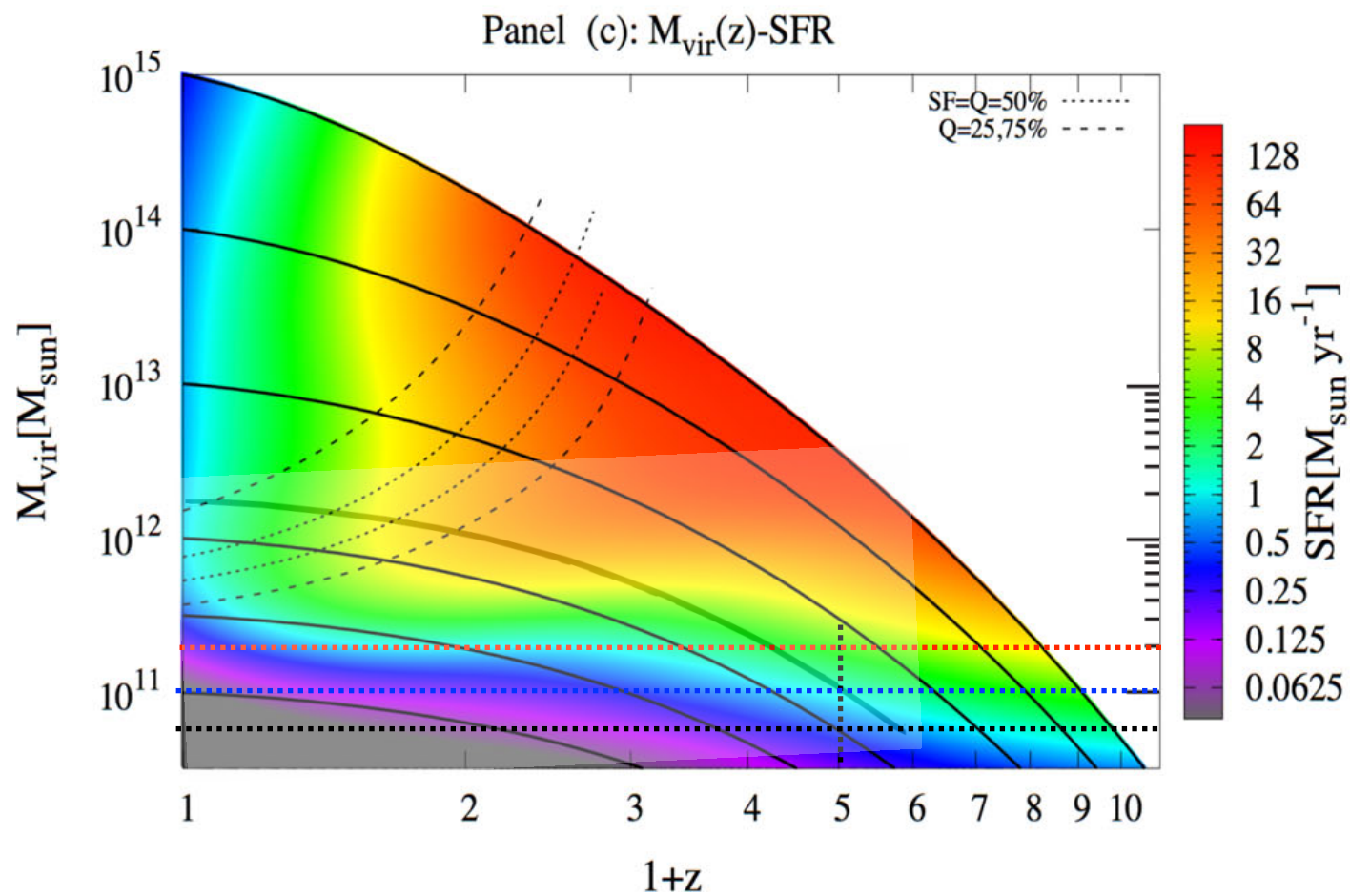
is about $(0.5-1) \times 10^9 M_{\odot}$

$M_{\text{vir}} = 0.5 \times 10^{11}$ at $z = 4$ for typical $10^{12} M_{\odot}$ halo at $z = 0$

$M_{\text{vir}} = 1 \times 10^{11}$ at $z = 4$ for typical $1.7 \times 10^{12} M_{\odot}$ halo at $z = 0$

$M_{\text{vir}} = 2 \times 10^{11}$ at $z = 4$ for AGORA $1.7 \times 10^{12} M_{\odot}$ halo at $z = 0$





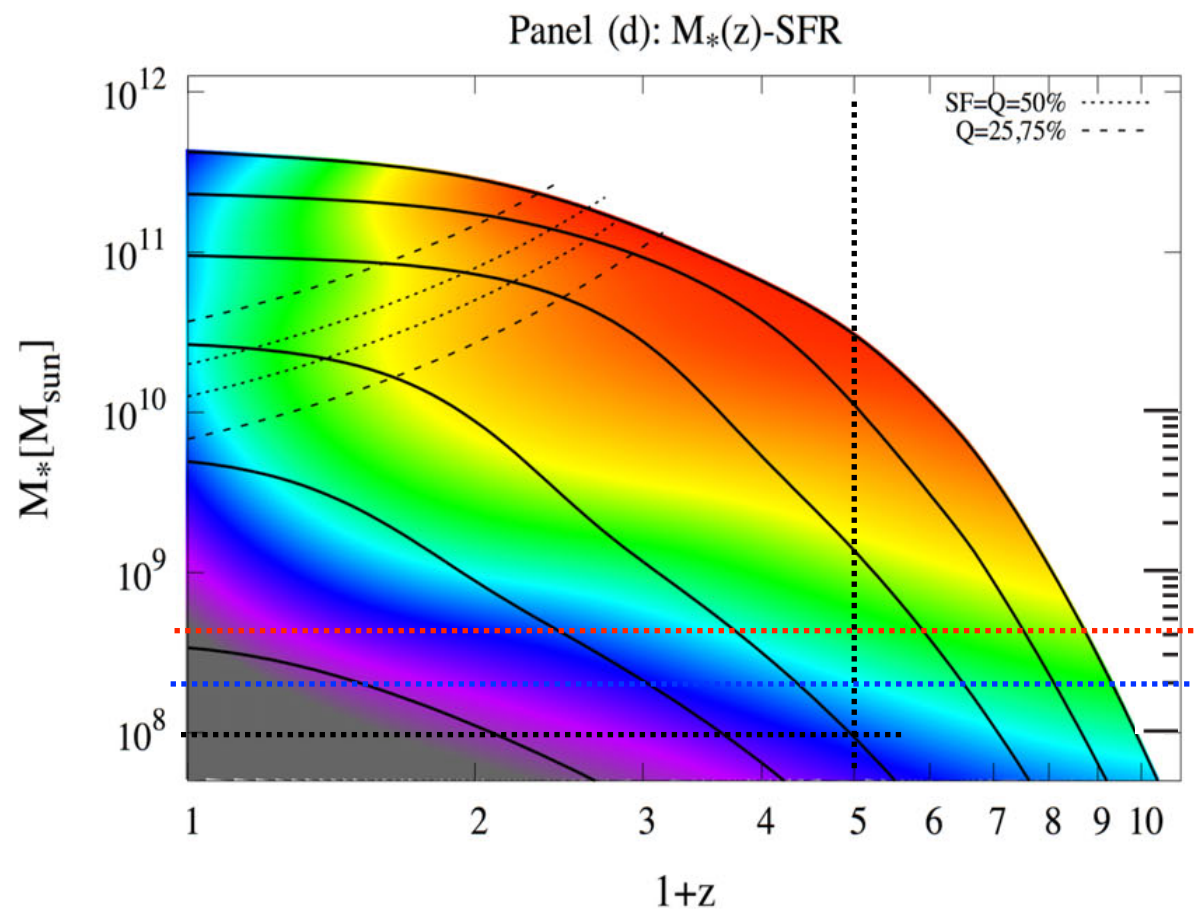
Constraining the galaxy–halo connection over the last 13.3 Gyr: star formation histories, galaxy mergers and structural properties

Aldo Rodríguez-Puebla, Joel R. Primack, Vladimir Avila-Reese, S. M. Faber 2017

$M_{\text{vir}} = 2 \times 10^{11}$ at $z = 4$ for AGORA $1.7 \times 10^{12} M_{\odot}$ halo at $z = 0$

$M_{\text{vir}} = 1 \times 10^{11}$ at $z = 4$ for typical $1.7 \times 10^{12} M_{\odot}$ halo at $z = 0$

$M_{\text{vir}} = 0.6 \times 10^{11}$ at $z = 4$ for typical $10^{12} M_{\odot}$ halo at $z = 0$



$M^*/M_{\odot}$ at $z = 4$

Rodriguez-Puebla+17 UnivMachine2018

typical 10^{12} halo at $z=0$	1×10^8	1×10^8
typical 1.7×10^{12} halo	2×10^8	3×10^8
AGORA 1.7×10^{12} halo	4×10^8	1×10^9

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$M_{\text{vir}} = 2 \times 10^{11}$ at $z = 4$ for AGORA $1.7 \times 10^{12} M_{\odot}$ halo at $z = 0$

$M_{\text{vir}} = 1 \times 10^{11}$ at $z = 4$ for typical $1.7 \times 10^{12} M_{\odot}$ halo at $z = 0$

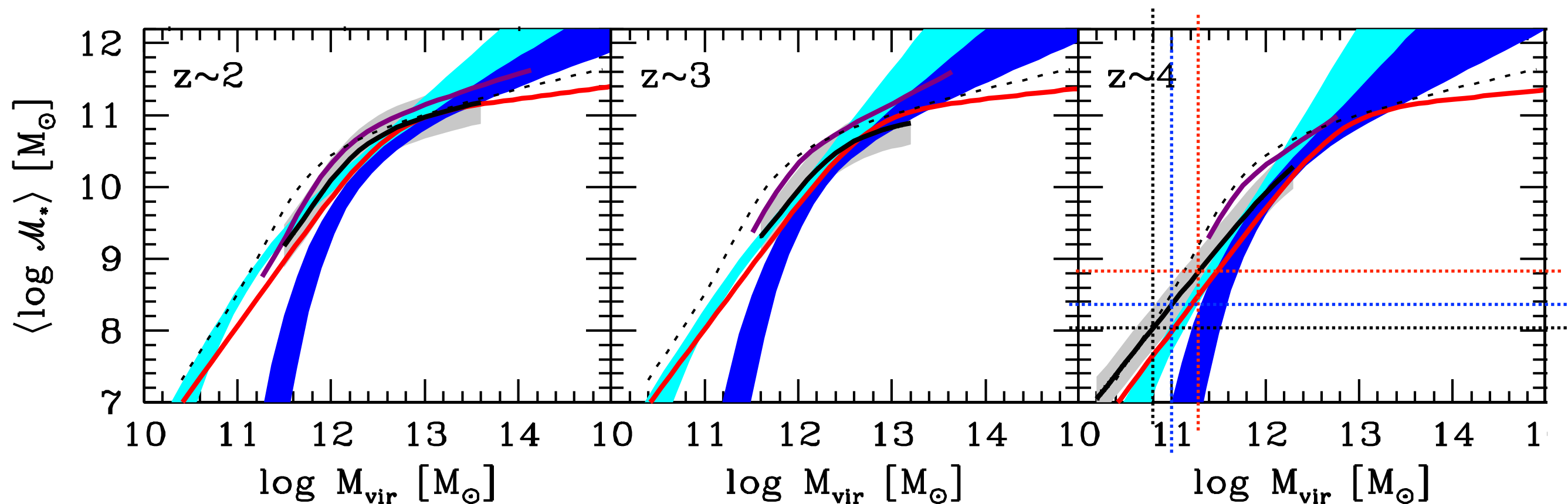
$M_{\text{vir}} = 0.6 \times 10^{11}$ at $z = 4$ for typical $10^{12} M_{\odot}$ halo at $z = 0$

$M^*/M_{\odot}$ at $z = 4$

Rodríguez-Puebla+17 UnivMachine2018

- This work
- Guo+2010
- Yang+2012 (SMF1)
- Yang+2012 (SMF2)
- Behroozi+2013
- Moster+2013

typical 10^{12} halo at $z=0$	1×10^8	1×10^8
typical 1.7×10^{12} halo	2×10^8	3×10^8
AGORA 1.7×10^{12} halo	6×10^8	1×10^9



UNIVERSEMACHINE: The Correlation between Galaxy Growth and Dark Matter Halo Assembly from $z = 0$ -10

Peter Behroozi, Risa H. Wechsler, Andrew P. Hearin, Charlie Conroy 2018

